

CHAPTER 6

Network Protocols

HTTP and HTTPS · FTP and TFTP · SMTP, POP3, IMAP · TELNET and SSH · PPP · VoIP · DNS · DHCP · SNMP, NTP, ICMP

QUICK TRICK

Learn the fast tricks first (details below):

1. **Web:** HTTP 80, HTTPS 443. **Mail:** SMTP 25 **sends**, POP3 110 and IMAP 143 **receive**. **Files:** FTP 21 (and 20), TFTP 69. **Login:** TELNET 23, SSH 22. **Names:** DNS 53. **Addresses:** DHCP 67/68.
2. **SMTP = push, POP3/IMAP = pull.** POP3 downloads (and usually deletes) mail; IMAP keeps it on the server.
3. **DHCP steps = DORA:** Discover, Offer, Request, Acknowledge.
4. **UDP** is used by DNS (queries), DHCP, TFTP, SNMP, NTP and VoIP voice. **TCP** is used by HTTP(S), FTP, SMTP, POP3, IMAP, TELNET, SSH.

6.1 What is a network protocol? The big map

● LEVEL 1 · BEGINNER

The problem. Two computers made by different companies must still understand each other. **The idea.** A **protocol** is a set of rules that fixes the **format** of messages, their **meaning** and their **order** (syntax, semantics, timing; Chapter 1). Most Internet services follow the **client-server model**: the client sends a **request**, the server sends a **response**, on a well-known **port** (Chapter 5).

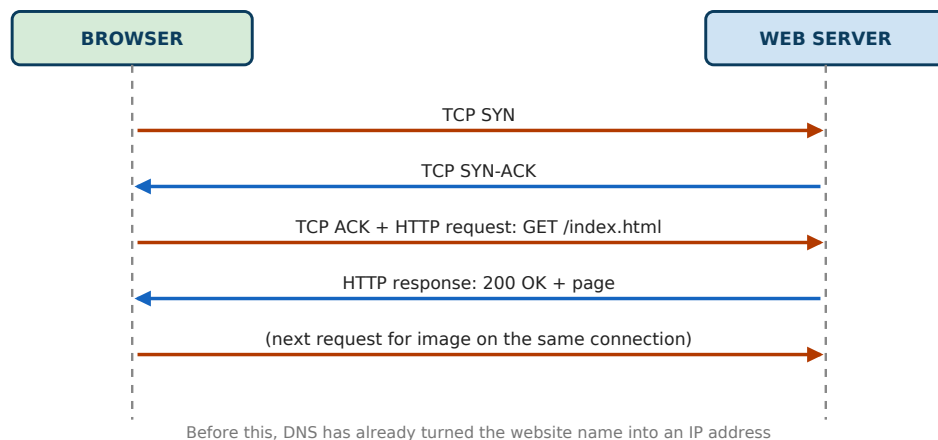
Analogy. Ordering food by phone: you greet, say the order, the shop repeats it, you pay, you say thanks. Each protocol is such a fixed routine for one service.

Protocol	Full form	Layer	Port	Transport	Purpose
HTTP	HyperText Transfer Protocol	Application	80	TCP	Web pages
HTTPS	HTTP Secure (HTTP over TLS)	Application	443	TCP	Secure web
FTP	File Transfer Protocol	Application	21 (control), 20 (data)	TCP	Upload / download files
TFTP	Trivial File Transfer Protocol	Application	69	UDP	Very simple file transfer
SMTP	Simple Mail Transfer Protocol	Application	25 (587, 465)	TCP	Send e-mail
POP3	Post Office Protocol version 3	Application	110 (995 secure)	TCP	Download e-mail
IMAP	Internet Message Access Protocol	Application	143 (993 secure)	TCP	Read e-mail on server
TELNET	Terminal Network	Application	23	TCP	Remote login (not secure)
SSH	Secure Shell	Application	22	TCP	Secure remote login
DNS	Domain Name System	Application	53	UDP (and TCP)	Name → IP address
DHCP	Dynamic Host Configuration Protocol	Application	67 (server), 68 (client)	UDP	Automatic IP settings
SNMP	Simple Network Management Protocol	Application	161, 162	UDP	Manage devices
NTP	Network Time Protocol	Application	123	UDP	Clock synchronisation
SIP	Session Initiation Protocol	Application	5060	UDP / TCP	VoIP call setup
PPP	Point-to-Point Protocol	Data Link	—	—	Direct link between 2 nodes
ICMP	Internet Control Message Protocol	Network	—	(IP itself)	Errors, ping

6.2 HTTP and HTTPS

● LEVEL 1 · BEGINNER

HTTP (HyperText Transfer Protocol) is the protocol of the **World Wide Web**. A **browser** (client) asks a **web server** for pages, images and files, and the server answers. It runs over **TCP, port 80**. HTTP is **stateless**: the server does not remember earlier requests; **cookies** are used to remember a user (login, shopping cart).



HTTP request	HTTP response
GET /index.html HTTP/1.1 (request line: method, resource, version) Host: www.example.com User-Agent: Chrome Accept: text/html (blank line) (body, only for POST / PUT)	HTTP/1.1 200 OK (status line: version, code, text) Content-Type: text/html Content-Length: 1256 (blank line) <html> ... page ... </html>

Method	Use
GET	Read (fetch) a resource; data goes in the URL
POST	Send data to the server (form, login); data goes in the body
PUT	Upload or replace a resource
DELETE	Remove a resource
HEAD	Like GET but only the headers, no body

Code class	Meaning	Examples
1xx	Informational	100 Continue
2xx	Success	200 OK, 201 Created
3xx	Redirection	301 Moved Permanently, 302 Found, 304 Not Modified
4xx	Client error	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found
5xx	Server error	500 Internal Server Error , 502 Bad Gateway, 503 Service Unavailable

Version	Feature
HTTP/1.0	Non-persistent: a new TCP connection for every object
HTTP/1.1	Persistent connections by default: many objects over one TCP connection; pipelining possible
HTTP/2 (2015)	Binary format, many requests multiplexed on one connection, header compression
HTTP/3	Runs over QUIC (built on UDP) for faster set-up and less blocking

Example (response time). A page has 1 base HTML file and 5 images. $RTT = 100\text{ ms}$; ignore file transmission time.

Non-persistent (one object after another, each needs a new TCP connection = 2 RTT): $6 \times 2\text{ RTT} = 12\text{ RTT} = \mathbf{1200\text{ ms}}$.

Persistent, no pipelining: $1\text{ RTT for TCP set-up} + 6 \times 1\text{ RTT} = 7\text{ RTT} = \mathbf{700\text{ ms}}$.

Persistent with pipelining: $1\text{ RTT (TCP)} + 1\text{ RTT (HTML)} + 1\text{ RTT (all 5 images together)} = 3\text{ RTT} = \mathbf{300\text{ ms}}$.

HTTPS = HTTP running over **TLS** (Transport Layer Security; its older name is SSL), **port 443**. It gives **encryption** (nobody can read the data), **integrity** (data is not changed) and **authentication** (the server proves its identity with

a **digital certificate**). The browser shows a padlock and the address starts with **https://**. Encryption and certificates are covered in the security chapter.

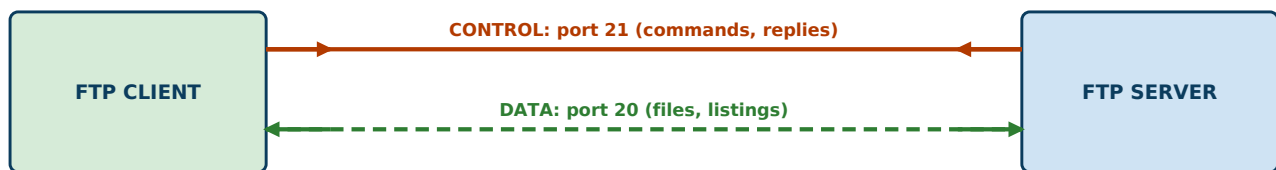
QUICK TRICK

HTTP is plain text, HTTPS is encrypted. Never type a password or card number on a page that is only HTTP. **404** = the page was not found (client side); **500** = the server itself failed.

6.3 FTP and TFTP

● LEVEL 1 · BEGINNER

FTP (File Transfer Protocol) uploads and downloads files between a client and a server. It uses **two TCP connections**:



Control information travels on its own connection: OUT-OF-BAND

Point	Detail
Control connection	Port 21. Opened first, stays open. Carries commands (USER, PASS, LIST, RETR, STOR, QUIT) and replies.
Data connection	Port 20 in active mode (server connects to the client). A new one is opened for each file or listing and closed after it.
Active vs passive	Active : the server opens the data connection. Passive : the client opens it to a port the server gives; works better through firewalls.
Modes of transfer	ASCII mode (text files) and binary (image) mode (programs, pictures, ZIP files)
Commands	RETR = download a file, STOR = upload a file, LIST = list files, CWD = change folder, PWD = print folder
Anonymous FTP	Public login (user "anonymous") for public downloads
Security	Passwords and data go as plain text , so it is not secure. Use SFTP (file transfer over SSH, port 22) or FTPS (FTP over TLS).

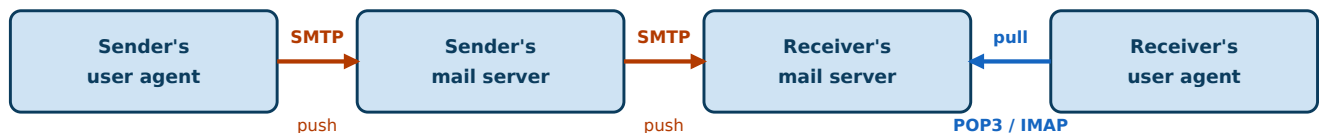
TFTP (Trivial File Transfer Protocol) is a very small version: it uses **UDP port 69**, has **no login, no directory listing** and only reads or writes a whole file. It is used to load boot files and device configurations on routers and switches inside a trusted LAN.

Point	FTP	TFTP
Transport	TCP (reliable)	UDP (simple, error checks done by TFTP itself)
Port	21 and 20	69
Authentication	Yes (user and password)	None
Features	Many commands, directories	Only get and put a file
Use	Website uploads, file servers	Network booting, device firmware

6.4 E-mail protocols: SMTP, POP3, IMAP

● LEVEL 1 · BEGINNER

E-mail uses several protocols together. The sender's program is a **user agent** (UA, for example Outlook or a mail app). Mail is stored in **mailboxes** on **mail servers** (also called **message transfer agents**). **MIME** (Multipurpose Internet Mail Extensions) lets mail carry attachments, pictures and non-English text, because SMTP itself only handles plain 7-bit ASCII text.



SMTP pushes mail out and between servers; POP3 or IMAP pulls it from the receiver's mailbox

Point	SMTP	POP3	IMAP
Full form	Simple Mail Transfer Protocol	Post Office Protocol v3	Internet Message Access Protocol
Job	Send mail (client → server, server → server)	Retrieve mail from the server	Retrieve and manage mail on the server
Direction	Push	Pull	Pull
Port	25 (server to server), 587 (client submission), 465 (SMTPS)	110 (995 with TLS)	143 (993 with TLS)
Mail stored	Only passes it on	Downloaded to one device; usually deleted from server	Stays on the server ; many devices see the same mailbox
Offline use	—	Good: read downloaded mail offline	Needs connection (can cache)
Folders	—	Basic (inbox only)	Full folders and flags on the server

A simple SMTP conversation

Client	Server reply	Meaning
(connects to port 25)	220 Service ready	Server is ready
HELO / EHLO client.example	250 OK	Introduce the client
MAIL FROM:<ravi@a.com>	250 OK	Who is sending
RCPT TO:<sita@b.com>	250 OK	Who should receive
DATA	354 Start mail input	Server waits for the message
(message, then a line with a single dot)	250 OK	Message accepted
QUIT	221 Closing connection	End of session

COMMON MISTAKE

- Saying SMTP is used to **read** mail. SMTP only **sends**; reading uses POP3 or IMAP (or a web browser with a webmail site, which uses HTTP).
- Mixing POP3 and IMAP: POP3 = **download and delete** (one device); IMAP = **keep on server** (many devices).

6.5 TELNET and SSH: remote login

● LEVEL 1 • BEGINNER

TELNET lets you log in to a distant computer and type commands as if you sat in front of it. It uses **TCP port 23** and a **Network Virtual Terminal (NVT)** so that different kinds of terminals can work together. TELNET sends **everything, including the password, as plain text**, so anyone listening can read it. It is therefore **insecure** and now used only for testing or old devices.

SSH (Secure Shell) does the same job but **encrypts** the whole session and uses **public-key authentication** (TCP port 22). It also carries secure file copy (SCP, SFTP) and tunnels other traffic.

Point	TELNET	SSH
Port	23	22
Encryption	None (plain text)	Yes
Authentication	Password in clear	Password or public key, server is verified
Use today	Rare, lab or old devices	Standard for servers and routers

6.6 PPP: Point-to-Point Protocol

● LEVEL 2 · EXAM

PPP is a **Data Link layer** protocol for a **direct link between two nodes**: dial-up modems, DSL (PPPoE), leased lines and routers. It is the protocol your old dial-up connection used to talk to the ISP.

Feature	Meaning
Framing	Packs data into frames with flag and delimiters
LCP	Link Control Protocol : sets up, configures, tests and closes the link
NCP	Network Control Protocol : one for each network-layer protocol (for IP it is IPCP), so PPP can carry IP and others
Authentication	PAP (Password Authentication Protocol): 2-way handshake, password sent in clear. CHAP (Challenge Handshake Authentication Protocol): 3-way handshake with a challenge, password never sent
Error detection	CRC (FCS) in the frame trailer. PPP does not correct errors or do flow control
Byte stuffing	Flag byte 0x7E inside data is replaced using escape byte 0x7D (on asynchronous links; on synchronous links bit stuffing like HDLC is used)

PPP phases:



Flag	Address	Control	Protocol	Data (payload)	FCS (CRC)	Flag
01111110	11111111	00000011	1-2 bytes	Up to a maximum size	2 or 4 bytes	01111110

QUICK TRICK

PPP = **LCP** to open the link, **Authenticate** (CHAP is safer than PAP), **NCP** to set up IP. The Protocol field says what is inside (for example IP).

6.7 VoIP: Voice over IP

● LEVEL 2 · EXAM

VoIP sends **voice calls as IP packets over the Internet** instead of using the telephone company's circuit-switched network (Chapter 4). Examples: WhatsApp calls, Skype, Google Meet, Zoom, IP phones. It saves cost, especially for long distance, and the same network carries voice, video and data.

Step	What happens
1. Digitise	Voice is sampled and coded by a codec (G.711 = 64 kbps, G.729 = 8 kbps)
2. Packetise	A short slice of voice (for example 20 ms) is put in a packet with RTP , UDP and IP headers
3. Signalling	SIP (port 5060) or H.323 finds the other person, rings, answers and ends the call
4. Transport	Voice packets travel by UDP ; RTP (Real-time Transport Protocol) adds sequence numbers and time stamps to play them in order
5. Gateway	A VoIP gateway links the Internet to ordinary phone lines (PSTN)

Why UDP, not TCP? A voice packet that arrives late is useless, so resending it (as TCP would) only adds delay. A few lost packets are better than a delayed call.

Quality problem	Meaning	Target / remedy
Latency (delay)	Time for voice to reach the other side	One-way delay should be below about 150 ms (ITU G.114)
Jitter	Variation of delay between packets	Jitter buffer at the receiver
Packet loss	Packets dropped on the way	Loss concealment, QoS priority for voice

Example (bandwidth). G.711 at 64 kbps, packets every 20 ms. Payload per packet = $64\,000 \times 0.02 \div 8 = 160$ bytes. Add RTP 12 + UDP 8 + IP 20 = 40 bytes, so each IP packet = **200 bytes**. There are 50 packets per second, so rate on the network = $200 \times 8 \times 50 = 80$ kbps (more than the 64 kbps of pure voice).

6.8 DNS: Domain Name System

● LEVEL 1 · BEGINNER

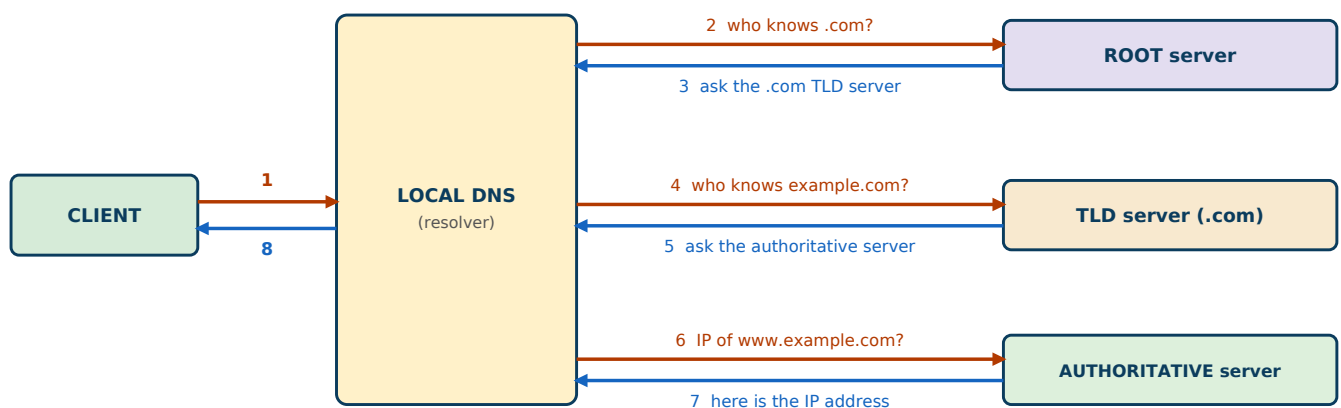
People remember names like **www.example.com**; computers need IP addresses like 93.184.216.34. **DNS** is the Internet's **phone book**: it translates a domain name into an IP address (and back). It is a **distributed, hierarchical database** spread over thousands of servers. DNS normally uses **UDP port 53** for queries; **TCP port 53** is used for big answers and zone transfers.

Domain name hierarchy. Read a name from right to left: **www . example . com .** The final dot is the **root**; **com** is the **top-level domain (TLD)**; **example** is the second-level domain; **www** is a host or subdomain. A name written in full is a **Fully Qualified Domain Name (FQDN)**.

Kind of TLD	Examples
Generic (gTLD)	.com, .org, .net, .edu, .gov, .info
Country code (ccTLD)	.in (India), .uk, .us, .jp, .au
India-specific second level	.co.in, .gov.in, .nic.in, .ac.in, .res.in

Name server	Role
Local (recursive) resolver	The ISP's or your own DNS server; takes the client's question, does the work and keeps a cache
Root server	Knows the servers of every TLD. There are 13 root server addresses (A to M) , each served by many machines worldwide
TLD server	Knows the authoritative servers for names under .com, .in and so on
Authoritative server	Holds the real records for the domain and gives the final answer

How a name is resolved (iterative query, empty cache)



Recursive vs iterative. The client asks its local resolver a **recursive** question ("get me the final answer"). The resolver then asks the other servers **iteratively** (each says "I do not know, ask that server") until it gets the IP address. The answer is stored in a **cache** for a time called **TTL** (time to live), so the next request is quick.

Record	Meaning	Example
A	Name → IPv4 address	www.example.com → 93.184.216.34
AAAA	Name → IPv6 address	www.example.com → 2606:2800:...
CNAME	Alias (canonical name) of another name	web.example.com → www.example.com
MX	Mail exchanger: the server that receives mail for the domain	example.com → mail.example.com
NS	Name server of the domain	example.com → ns1.example.com
PTR	IP address → name (reverse lookup)	34.216.184.93 → example.com
SOA	Start of authority: zone information	—

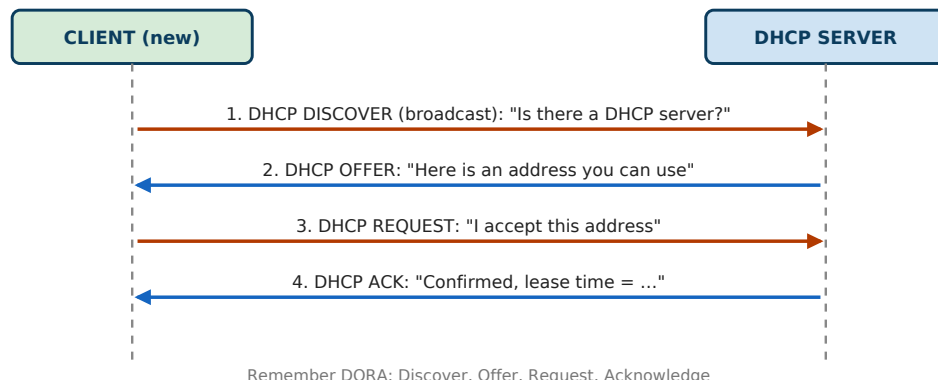
QUICK TRICK

Resolution order: **local cache** → **resolver** → **root** → **TLD** → **authoritative**. Tools to test DNS: **nslookup** and **dig**. DNS is needed before almost every HTTP request.

6.9 DHCP: Dynamic Host Configuration Protocol

● LEVEL 2 · EXAM

DHCP gives a computer its network settings **automatically**: **IP address, subnet mask, default gateway and DNS server**. Without it an administrator would have to type these on every device. A DHCP server hands out an address from a pool for a limited time, called a **lease**. It uses **UDP: port 67 (server) and port 68 (client)**. Home routers have a DHCP server built in.



Remember DORA: Discover, Offer, Request, Acknowledge

Lease timers. The client tries to **renew** at 50% of the lease time (T1) and, if that fails, **rebinds** with any server at 87.5% (T2). **Example:** lease = 8 hours → renew at 4 hours, rebind at 7 hours.

APIPA. If a Windows PC finds no DHCP server, it gives itself an address in the range **169.254.x.x** (Automatic Private IP Addressing). Seeing 169.254.x.x usually means "DHCP failed".

Static IP	Dynamic IP (DHCP)
Fixed, typed by the administrator; for servers, printers, routers	Given automatically; may change after the lease; for ordinary hosts

6.10 SNMP, NTP and ICMP

● LEVEL 2 · EXAM

Protocol	Job	How it works
SNMP	Monitors and manages devices (routers, switches, printers)	A manager asks agents on devices for values stored in a MIB (Management Information Base). UDP 161; traps from agents to UDP 162.
NTP	Keeps clocks of computers the same	Gets accurate time from time servers, UDP 123
ICMP	Reports errors and tests reachability	ping sends Echo request and gets Echo reply; traceroute shows each router on the path using the TTL field. ICMP is carried inside IP (not TCP or UDP).
ARP	Finds the MAC address for an IP address on the local network	Broadcasts "Who has this IP?"; the owner replies with its MAC

6.11 Common mistakes

● LEVEL 2 · EXAM

COMMON MISTAKE

- Saying HTTP uses UDP. HTTP and HTTPS use **TCP** (HTTP/3 over QUIC is the one exception).
- Mixing the ports: HTTP 80 / HTTPS 443, SMTP 25 / POP3 110 / IMAP 143, FTP 21 / TELNET 23 / SSH 22.
- Saying FTP uses one connection. It uses **two**: control (21) and data (20).
- Saying SMTP downloads mail. SMTP **sends** only.
- Believing DNS is a single computer. It is a worldwide **distributed hierarchy**.
- Saying TELNET is secure. It is **plain text**; use SSH.
- Saying PPP is a network-layer protocol. PPP is a **data link** layer protocol.

Quick-Solve Tricks — Summary

Situation	Trick
Web	HTTP 80, HTTPS 443 (TLS); stateless; 404 not found, 500 server error
HTTP/1.1	Persistent connections; time = 1 RTT (TCP) + 1 RTT per object (no pipelining)
File transfer	FTP: control 21 + data 20, TCP · TFTP: 69, UDP, no login
E-mail	SMTP sends (25/587) · POP3 downloads (110) · IMAP keeps on server (143)
Remote login	TELNET 23 plain text · SSH 22 encrypted
PPP	Data link; LCP + NCP; PAP (clear) vs CHAP (challenge)
VoIP	Codec + RTP over UDP + SIP signalling; delay below 150 ms; G.711 = 64 kbps
DNS	Name → IP; UDP 53; root → TLD → authoritative; A, AAAA, CNAME, MX, NS, PTR
DHCP	DORA; UDP 67 / 68; renew at 50%, rebind at 87.5%; APIPA 169.254.x.x
Other	SNMP 161/162 · NTP 123 · ICMP = ping, traceroute · ARP = IP → MAC

BPSC QUICK CHECK — Chapter 6

HTTP → port 80, TCP	HTTPS → port 443, TLS encryption
FTP → ports 21 and 20	TFTP → port 69, UDP
SMTP → send mail	POP3 / IMAP → receive mail
TELNET → insecure, 23	SSH → secure, 22
DNS → name to IP	DHCP → automatic IP (DORA)
PPP → data link, LCP and NCP	VoIP → voice over UDP with RTP

Traps: SMTP vs POP3 vs IMAP · active vs passive FTP · TELNET vs SSH · PAP vs CHAP · recursive vs iterative DNS · static vs dynamic IP.

Practice questions

● LEVEL 1 · BEGINNER

- **Q1.** Expand HTTP, FTP, SMTP, POP3 and DNS.
- **Q2.** Which protocol is used to send e-mail?
- **Q3.** Which protocol gives secure web browsing, and what is its port?
- **Q4.** Which protocol translates a domain name into an IP address?
- **Q5.** Which protocol is used for remote login but is not secure?
- **Q6.** What does VoIP do?

● LEVEL 2 · EXAM

- **Q7.** Give three differences between POP3 and IMAP.
- **Q8.** Why does FTP use two connections?
- **Q9.** What do the HTTP status codes 404 and 500 mean?
- **Q10.** Name the four steps of DHCP in order.
- **Q11.** At which layer does PPP work, and what are LCP and NCP?
- **Q12.** Why does VoIP use UDP and not TCP?

● LEVEL 3 · BPSC / PGT

- **Q13.** A page has 1 base HTML file and 5 images, $RTT = 100$ ms (ignore transmission time). Find the total time for (a) non-persistent HTTP one object at a time, (b) persistent HTTP without pipelining, (c) persistent HTTP with pipelining.
- **Q14.** G.711 voice (64 kbps) is sent in 20 ms packets with RTP (12), UDP (8) and IP (20) headers. Find the payload size, the IP packet size and the bit rate on the network.
- **Q15.** A DHCP lease is 8 hours. When does the client try to renew and when does it rebind?
- **Q16.** Describe how a resolver with an empty cache finds the IP of `www.example.com`.
- **Q17.** Give the protocol and port for: (a) browsing a secure site, (b) a mail client submitting a message, (c) reading mail while leaving it on the server, (d) logging in securely to a server, (e) a DNS lookup.
- **Q18.** Compare PAP and CHAP. Why is CHAP better?

Answers

Q	Solution
1	HyperText Transfer Protocol; File Transfer Protocol; Simple Mail Transfer Protocol; Post Office Protocol version 3; Domain Name System.
2	SMTP.
3	HTTPS (HTTP over TLS), port 443.
4	DNS.
5	TELNET (port 23, plain text).
6	Sends voice calls as IP packets over the Internet or any IP network.
7	POP3: downloads mail and usually deletes it from the server; mainly one device; port 110. IMAP: leaves mail on the server; many devices see the same mailbox; supports folders; port 143.
8	One connection (port 21) stays open for commands and replies, and a separate data connection (port 20 in active mode) carries each file. This keeps control information separate from data (out-of-band).
9	404 = Not Found (the resource does not exist, client side). 500 = Internal Server Error (the server failed, server side).
10	Discover, Offer, Request, Acknowledge (DORA).
11	Data Link layer. LCP (Link Control Protocol) sets up, tests and closes the link. NCP (Network Control Protocol) configures each network-layer protocol, for example IPCP for IP.
12	Late voice packets are useless, so retransmission by TCP only adds delay. UDP is faster and a few lost packets are acceptable.
13	(a) $6 \text{ objects} \times 2 \text{ RTT} = 12 \text{ RTT} = \mathbf{1200 \text{ ms}}$. (b) $1 + 6 = 7 \text{ RTT} = \mathbf{700 \text{ ms}}$. (c) $1 + 1 + 1 = 3 \text{ RTT} = \mathbf{300 \text{ ms}}$.
14	Payload = $64\,000 \times 0.02 \div 8 = \mathbf{160 \text{ bytes}}$. IP packet = $160 + 12 + 8 + 20 = \mathbf{200 \text{ bytes}}$. Rate = $200 \times 8 \times 50 \text{ packets/s} = \mathbf{80 \text{ kbps}}$.
15	Renew at 50% = 4 hours . Rebind at 87.5% = $0.875 \times 8 = \mathbf{7 \text{ hours}}$.
16	The client asks the local resolver (recursive). The resolver asks a root server, which refers it to the .com TLD server; the TLD server refers it to the authoritative server of example.com; that server returns the IP. The resolver caches it and answers the client.
17	(a) HTTPS, 443. (b) SMTP, 587 (or 25). (c) IMAP, 143. (d) SSH, 22. (e) DNS, UDP 53.
18	PAP: 2-way handshake, the password is sent in clear text, checked once. CHAP: 3-way handshake; the server sends a challenge, the client replies with a hash of the challenge and secret, so the password is never sent and the challenge can be repeated during the session.

REMEMBER

Next: Chapter 7 — IP Addressing and Subnetting (IPv4 classes, private and public addresses, subnet masks, CIDR, subnetting and VLSM, IPv6, NAT, routing basics and routing protocols).

Roadmap: Ch 8 Web Services (WWW, HTML, XML, URL, domain names, hosting, web browsers and servers) · Ch 9 Network Security · Ch 10 Mobile Telecom, Cloud and IoT.